

⑤ $mBnL$
 $\downarrow \quad \downarrow$
 m Binary bits n pulses
 L : level of pulses
 $\begin{cases} B: \text{Binary (2)} \\ T: \text{Ternary (3)} \\ Q: \text{Quaternary (4)} \end{cases}$

$$\text{coding rate} = \frac{m}{n}$$

$$2^m \leq L^n$$

E.g. 2B1Q

$$\downarrow$$

$$2^2 \leq 4^1$$

$$\downarrow$$

coding rate = 2 bits/pulse

4B3T

$$\downarrow$$

$$2^4 \leq 3^3$$

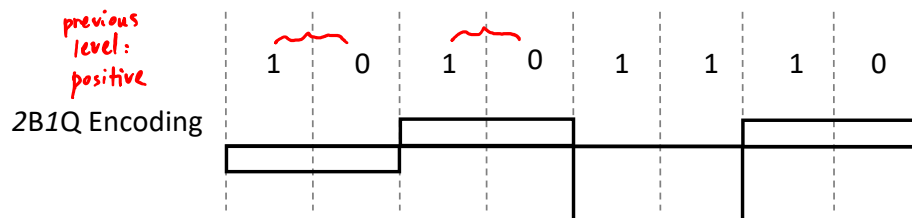
5B3T is not possible

$$2^5 \leq 3^3$$

$$>$$

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2B1Q Coding* & mBnL Coding



2B1Q $\pm \frac{A}{2}, \pm \frac{3A}{2}$

	Previous level: positive "1"	Previous level: negative
Next bits	Next level	Next level
→ "00"	+A/2	-A/2
"01"	+3A/2	-3A/2
"10"	-A/2 ✓	+A/2 ✓ ✓
"11"	-3A/2 ✓	+3A/2

mBnL

- ⊕ Maps block of m information bits into n pulses
- ⊕ There is a total of L different levels of pulses

* This version of 2B1Q coding is based on "Data Communications and Networking" by Behrouz A. Forouzan.

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2B1Q:

$$\text{bit rate} = \text{coding rate} * \text{Baud rate}$$

$$\text{e.g. } 10 \text{ Mbps} = 2 * 5 \text{ MHz}$$

Application : bandwidth-limited system

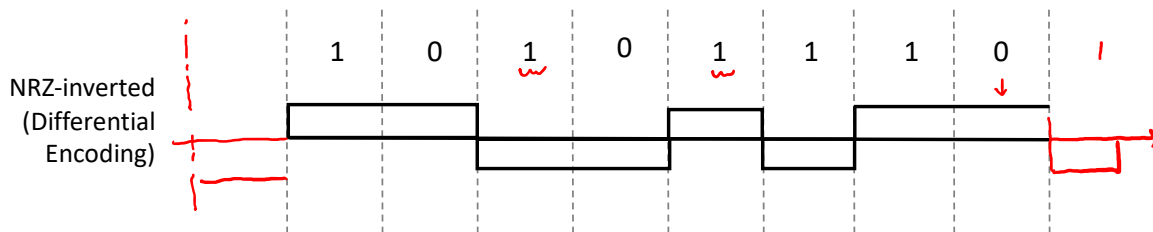
DSL

{ Digital
Subscriber
Line

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⑥

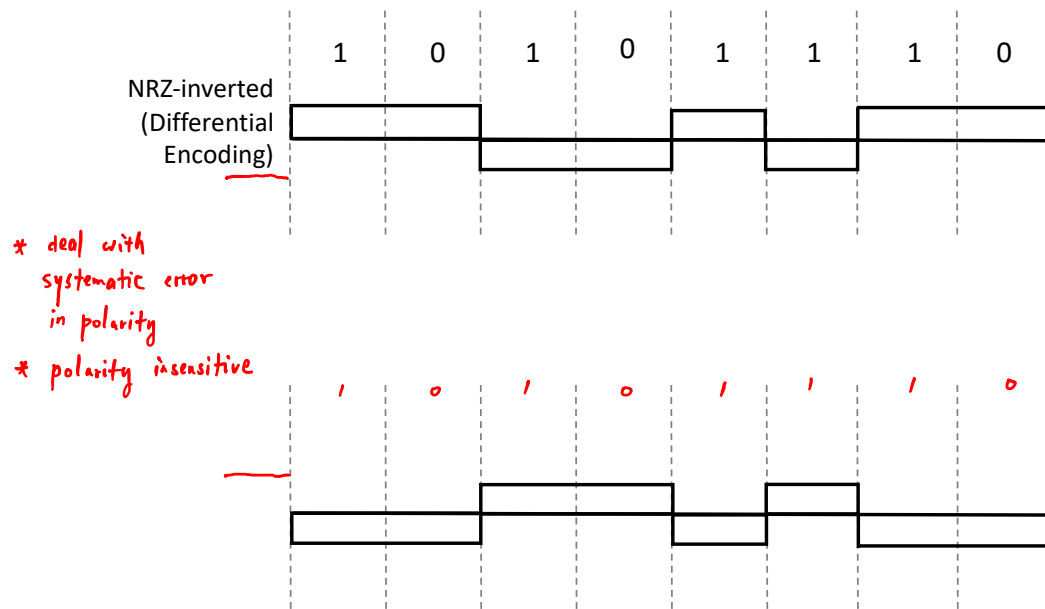
NRZ-Inverted Coding



- Two signal formats: $\{-A/2, +A/2\}$ *
- "1" maps to transition in signal format at beginning of the bit interval
- "0" maps to no transition
- Differential line coding { reference signal format
relation between adjacent signal formats
- Errors occur in pairs *
- Long string of "0"s causes receiver to lose synch

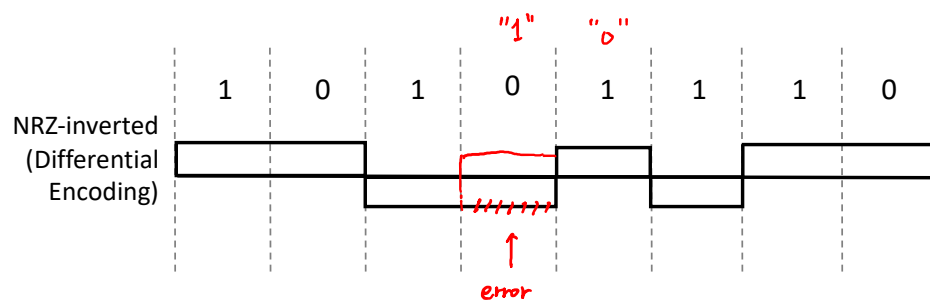
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NRZ-Inverted Coding



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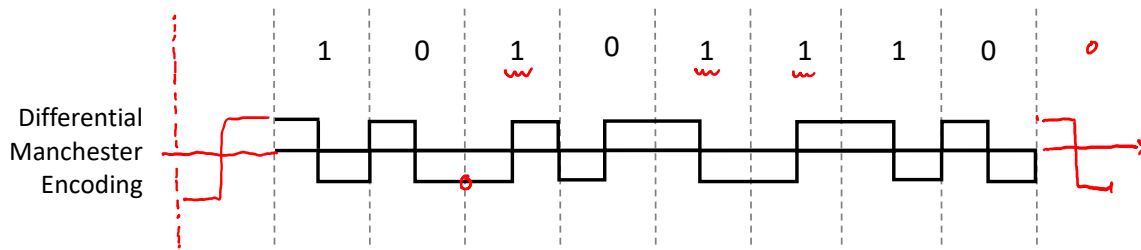
NRZ-Inverted Coding



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Differential Manchester Coding



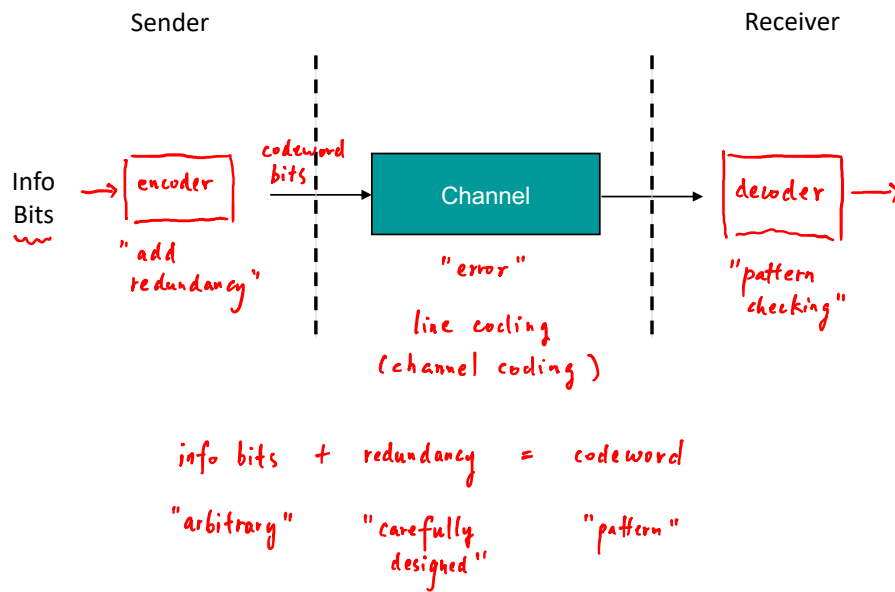
- ⊕ Systematic error in polarity (i.e., + become – and vice versa) is possible
 - ➡ Manchester Coding can not handle this type of error
- ⊕ Differential Manchester Coding provides robustness to this type of error
 - ➡ “1” maps to transition in signal format at beginning of the bit interval
 - ➡ “0” maps to no transition
 - ➡ Another type of differential line coding
 - ➡ Errors occur in pairs



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Topic 3: Error Detection and Recovery

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General Error Detection System

- ⊕ Transmitter (encoder) **adds redundancy** to user **information** to become **codewords** and transmit codewords over communication channel
- ⊕ All transmitted codewords satisfy **certain pattern** that is agreed upon between transmitter and receiver
- ⊕ If a received codeword doesn't satisfy the pattern, it is in error
 - ➡ Error detected!



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Example: Single Even Parity Check Code

- Append an overall parity check bit to k information bits

k Information Bits: $(i_{k-1}, \dots, i_1, i_0)$

$n-k=1$ Parity Check Bit: $C = (i_{k-1} + \dots + i_1 + i_0) \bmod 2$
 (redundancy) $\begin{cases} \text{even \# 1's in info bits} \\ \Rightarrow C = 0 \\ \text{odd \# 1's in info bits} \\ \Rightarrow C = 1 \end{cases}$

$n=k+1$ Codeword: $(i_{k-1}, \dots, i_1, i_0, C)$:
 $(k+1)$ bits pattern:
 even # 1's in all codewords

k : # info bits

$n-k$: # check/redundancy bits

n : # codeword bits ($n > k$)

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E.g. $k=4, n-k=1, n=5$

$\Rightarrow \left[\begin{array}{cccc|c} 1 & 0 & 1 & 1 & 1 \end{array} \right]_{1 \times 5}$
 info bits check bit

Error: $\checkmark \checkmark \times \times \checkmark$

$\left[\begin{array}{cccc|c} 1 & 0 & 0 & 0 & 1 \end{array} \right]_{1 \times 5} \Rightarrow \text{decoder}$

undetectable error

2-bit error

$M=2$

(M : # bit errors)

total # possible 2-bit errors?

$$\binom{5}{2} = \frac{5 \times 4}{2} = 10$$

$\times \checkmark \times \checkmark \times$

$\left[\begin{array}{ccccc} 0 & 0 & 0 & 1 & 0 \end{array} \right]_{1 \times 5}$

detectable error

3-bit error

$M=3$

$$\binom{5}{3} = 10$$

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